

Answer all questions.

- 1 The base units of the SI system include those of The base units of the SI system include those of

mass, kg; length, m; time, s; electric current, A.

Which base units would be needed to express the SI unit of potential difference (the volt)?

- A** m and A only.
B s and A only.
C m, s and A only.
D kg, m, s and A.

Ans: D

$$\left[V = \frac{W}{Q} = \frac{Fs}{It} = \frac{mas}{It} \right]$$
$$= \frac{\text{kg ms}^{-2} \text{ m}}{\text{A s}}$$

- 2 A car travelling at a speed of 15.0 m s^{-1} can be brought to rest in 1.20 s when a uniform braking